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UNIVERSITI
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ICT602

MOBILE TECHNOLOGY AND DEVELOPMENT

CHAPTER 3 - MOBILE DESIGN PRINCIPLES (3/9)



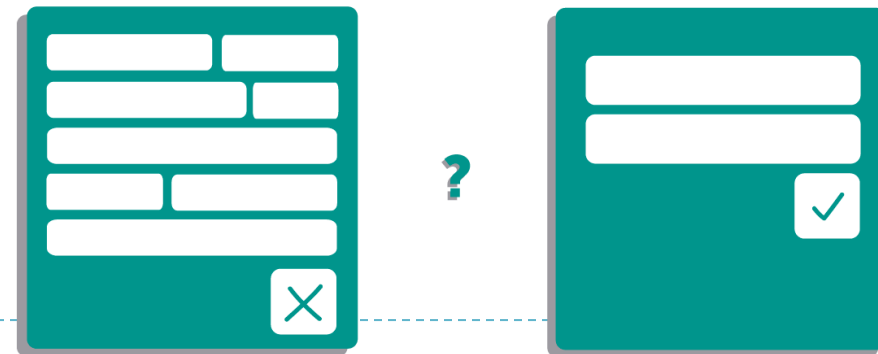
MOBILE UI AND DESIGN PATTERNS



UI Design Patterns



- ▶ A **design pattern documents** known good solutions to frequently occurring design problems.
- ▶ In most cases, the patterns themselves become encoded as user expectations.
- ▶ An application that **VIOLATES common design patterns** could jar and negatively impacts user expectation



Desktop vs. mobile app design: the big differences

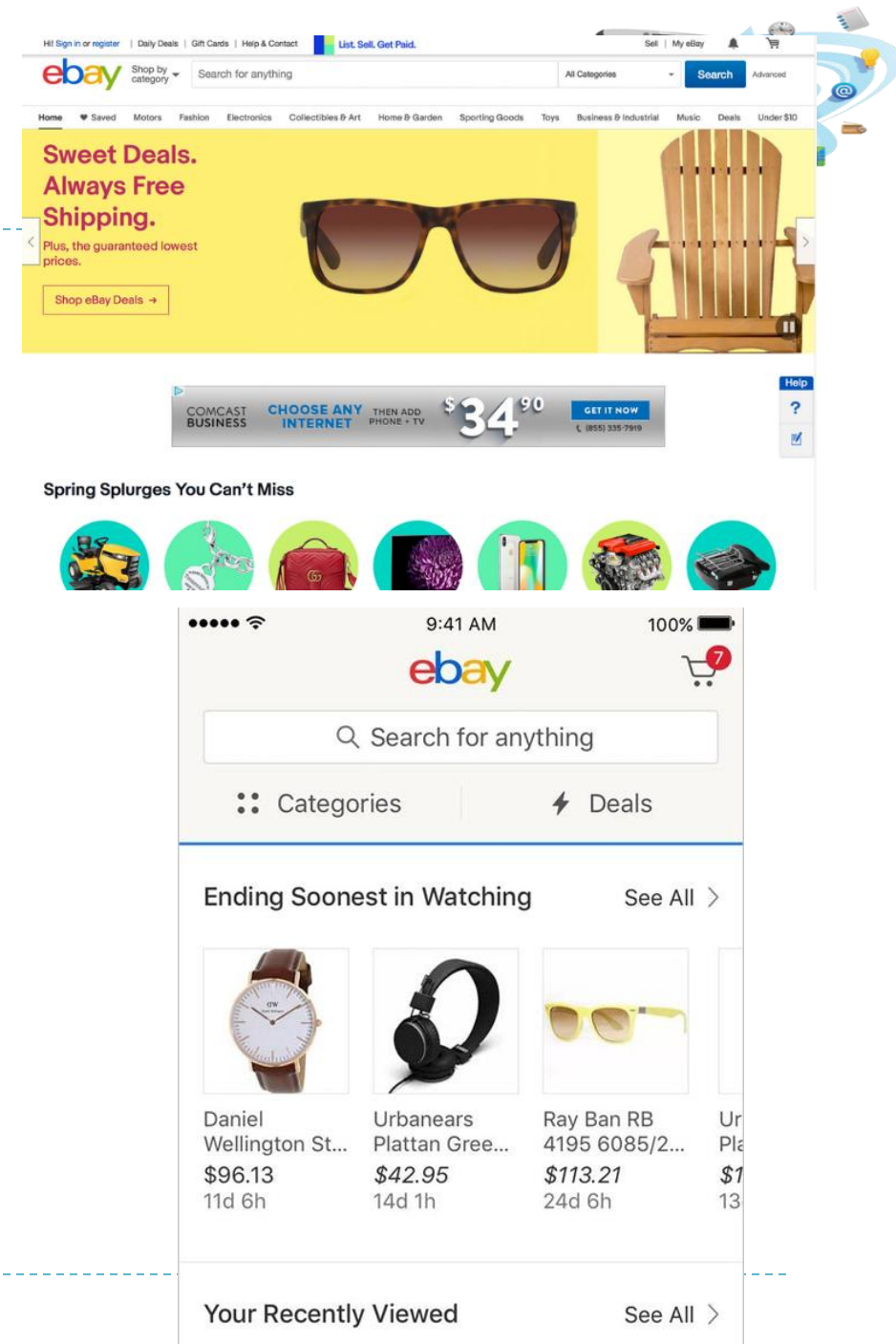
► Screen Size

► Desktop = large

- lot more than the opportunity to add more stuff.
- Screen size affects all aspects of design, especially navigation.
- Desktop apps can support fixed navigation bars

► Mobile = small

- Mobile apps must conserve screen space everywhere they can, so you must be aware of which elements are important enough to show.
- mobile are generally limited to pull-out menus



Desktop vs. mobile app design: the big differences



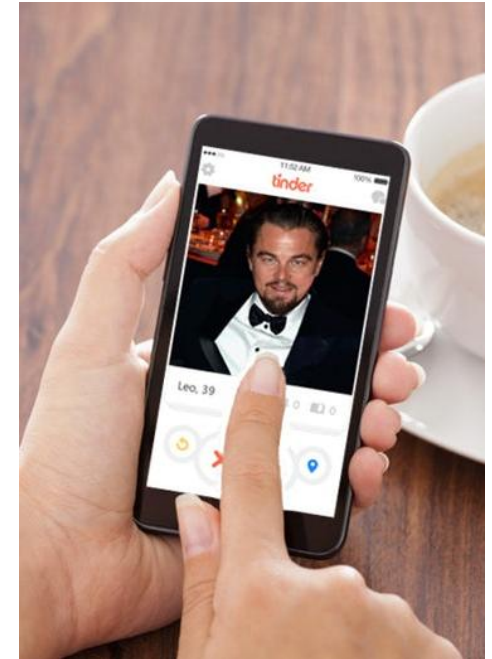
► Interaction

► Desktop = cursors

- Desktop apps can make full use of cursor interactivity: things like hover text or cursor-triggered animations.
- This allows desktop apps to feature entire screens full of pictures, with descriptive text only appearing on hover.

► Mobile = gestures

- can't hover or rollover on mobile apps
- infinite slate of gestures literally at your fingertips
- Can you imagine how boring Tinder would be if you had to click?



Desktop vs. mobile app design: the big differences



▶ **Organizing content**

▶ Desktop = columns

- ▶ Any content on desktop can appear in a traditional multi-column format—just like print content in newspapers and magazines.

▶ Mobile = scrolling

- ▶ Once content reaches a certain length, mobile apps need to use long scrolling.
- ▶ Mobile users actually prefer continuous scrolling, and this technique conserves screen space while making interaction more fun with gestures.

▶ Mobile = portrait and landscape

- ▶ Unlike desktop screens, mobile apps can switch between portrait and landscape views at will.

Desktop vs. mobile app design: the big differences



▶ **Functionality**

▶ Desktop = big tasks

- ▶ user prefer desktop apps for longer, more involved tasks.
- ▶ When longer tasks come up, users would rather find a seat and settle in to desktop apps with more functions, more content, and more special features.

▶ Mobile = experimental

- ▶ Mobile works for short spurts and quick tasks that come up in the moment
- ▶ What mobile lacks in functionality, it makes up for with ingenuity. Right now, mobile design is at the forefront of technology and offers tons of exclusive features that desktop just can't do.

Mobile UI Design principles



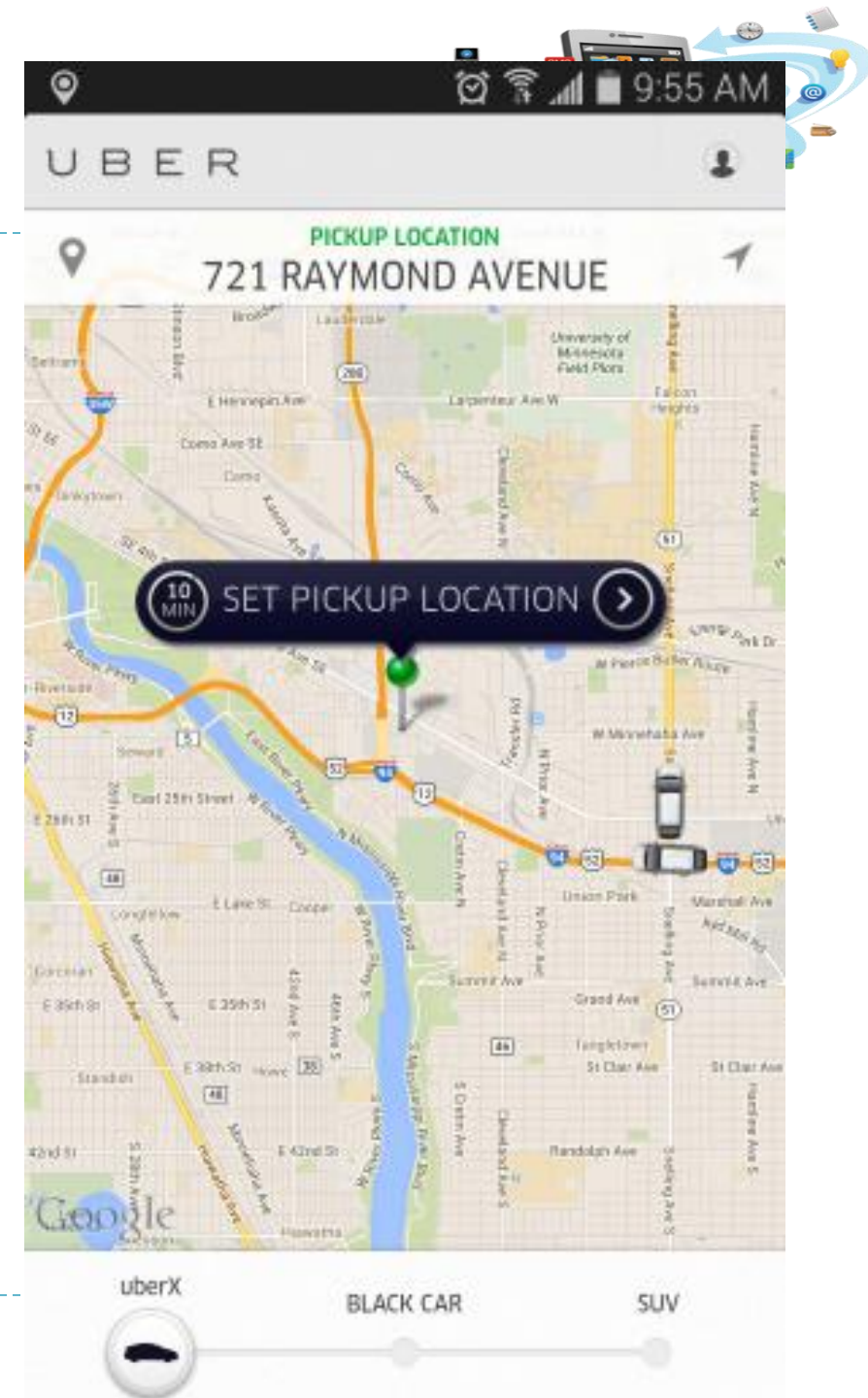
- ▶ In order to be useful, mobile apps should be **user-centric**.
- ▶ To be effective using an interface you've designed, people must be able to recognize **what it's for and how to use it**. There is simply **no room for confusion**.



Mobile UI Design principles

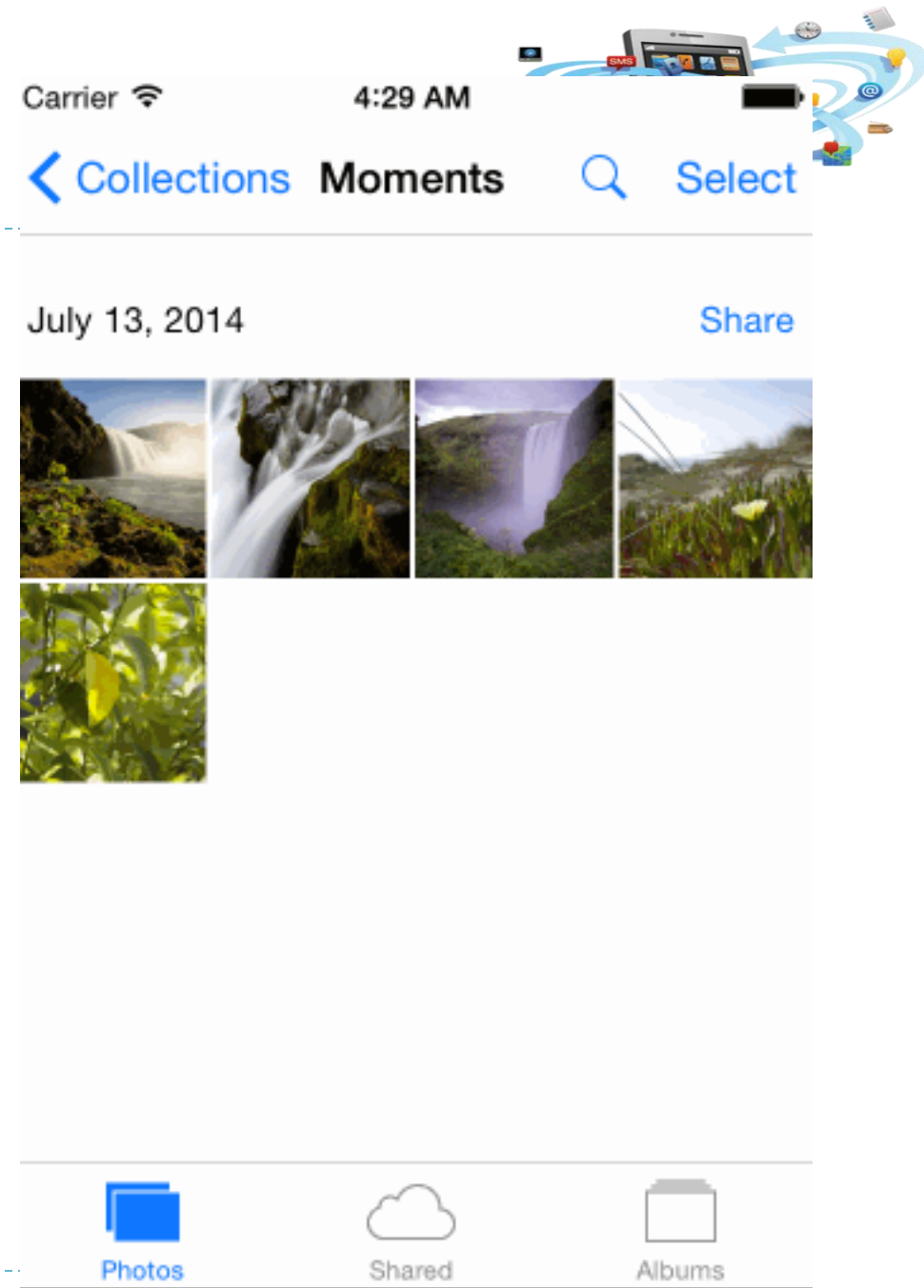
► one primary action per screen.

- One hundred clear screens is preferable to a single cluttered one.
- Example: Uber - Uber knows that the goal of the user who uses the app is to take a cab. The app does not overwhelm the user with other information: it automatically detects users location based on GEO data and the only thing users have to do is to select a pickup location



Make Navigation Self-Evident

- ▶ Mobile navigation must be **coherent**.
- ▶ Mobile navigation must be **consistent** for the app.
- ▶ Mobile navigation should **communicate the current location**.



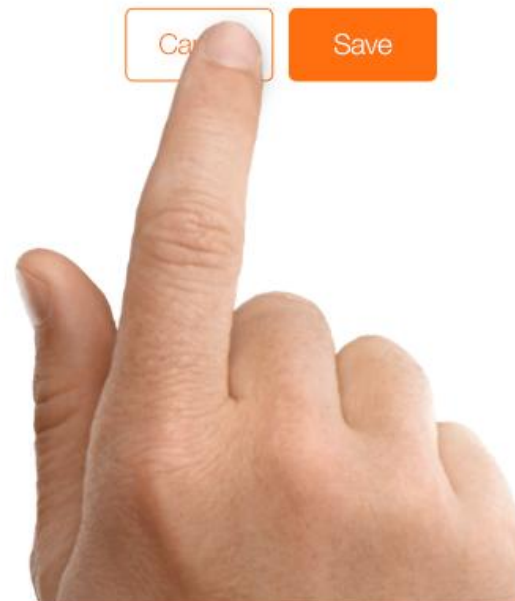


Design Finger-friendly Tap-targets

- ▶ When you're designing mobile interfaces, it's best to make your targets big enough so that they're easy for users to tap.
- ▶ The edges of the target are visible when the user taps it. This provides them with clear visual feedback that they're hitting the target accurately.



Don't.



Do.

Text Content Should Be Legible



- ▶ When compared with desktops, smartphones have relatively small screens,
 - ▶ one of the challenges of mobile design is to fit a lot of information on a small UI.
- ▶ A rule of thumb for mobile: text should be at least 11 points so it's legible at a typical viewing distance without zooming.

Heading

Sub-Headline

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Make Interface Elements Clearly Visible



- ▶ You should use *color* and *contrast* to help users see and interpret your content.
- ▶ Choose primary, secondary, and accent colors for your app that support usability. Ensure sufficient color contrast between elements so that users with low vision can see and use your app.
- ▶ Make sure there is ample contrast between the font color and the background so text are legible

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Minimize Need For Typing



- ▶ Typing on a mobile is a slow and error-prone process. It's therefore best to always try to minimize the amount of typing required to use a mobile app.
- ▶ Keep forms as short and simple as possible by removing any unnecessary fields.

"This is going to be painful."

"This will be a breeze."



Examples of UI Design Patterns Gallery

- ▶ Android UI Pattern document for designing Notifications

<https://developer.android.com/design/patterns/notifications.html>

Other Mobile UI Design Pattern galleries

- ▶ <http://www.mobile-patterns.com/>
- ▶ <http://android.inspired-ui.com/>



MOBILE UI DESIGN ELEMENT



Elements of Mobile UI Design



▶ Screen Design

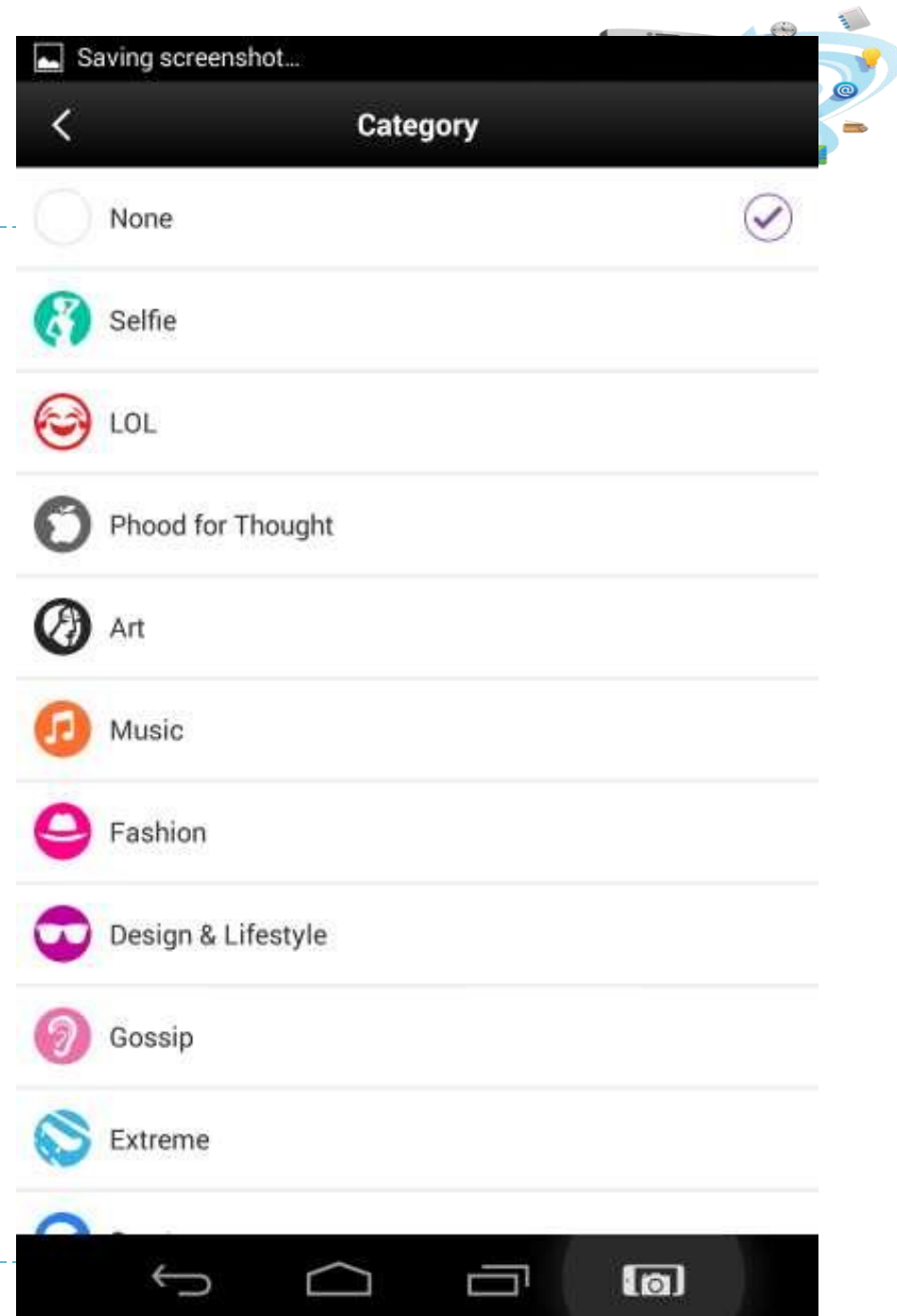
- ▶ List-based Layout
- ▶ Grid-Based Layout

▶ Application Navigation

- ▶ Tab Navigation
- ▶ ViewPager Navigation
- ▶ Navigation Drawer
- ▶ Dropdown Navigation

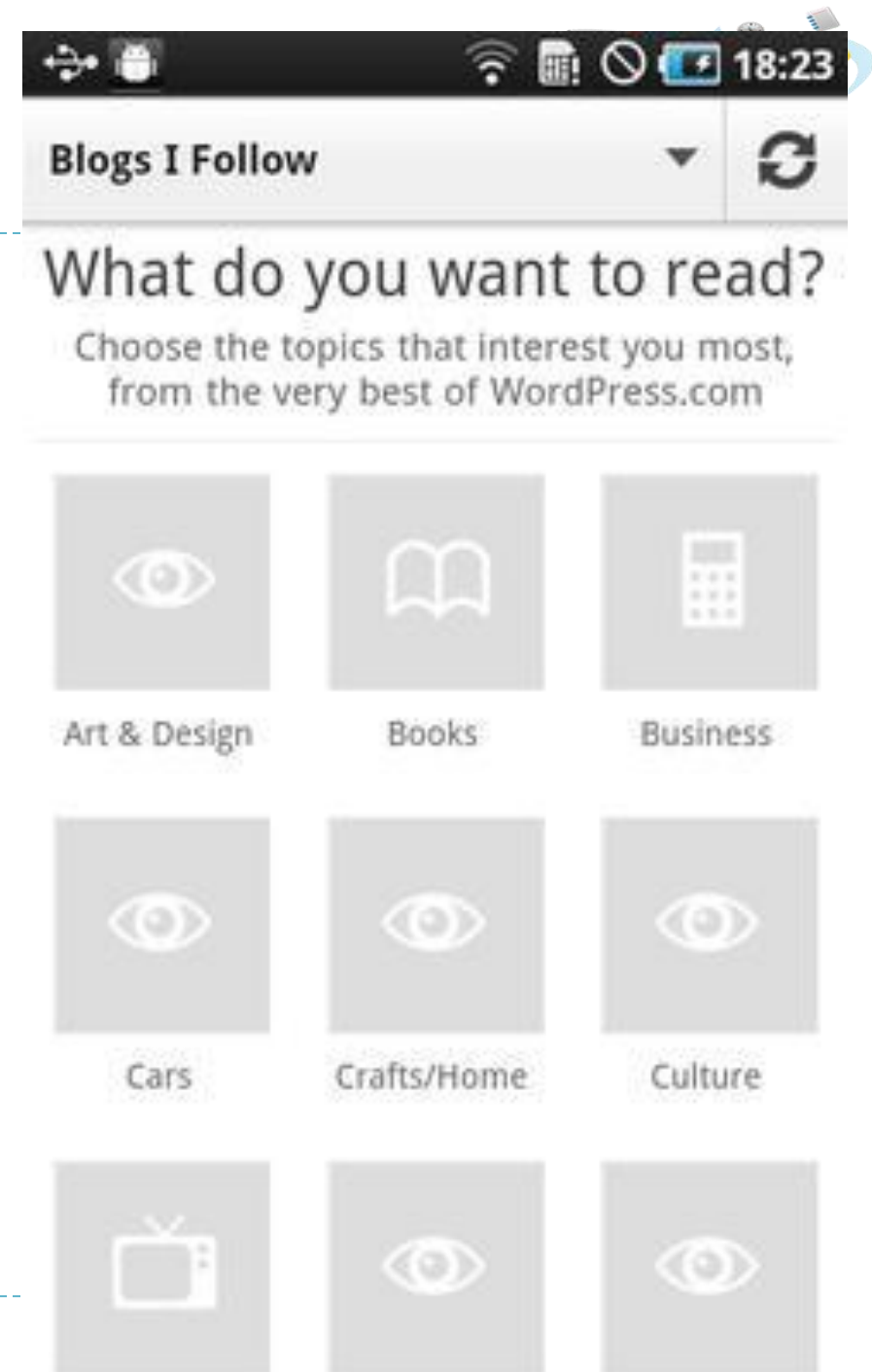
Screen Design: List-based Layout

- ▶ The **most common** screen in any mobile application designed to display **a lot of data**
- ▶ Rationale for using List-based Layout:
 - ▶ Most mobile phones are oriented **vertically**, with screens taller than they are wide
 - ▶ List is more natural to use by users as they can **scroll down** to search for data more efficiently,



Screen Design: Grid based Layout

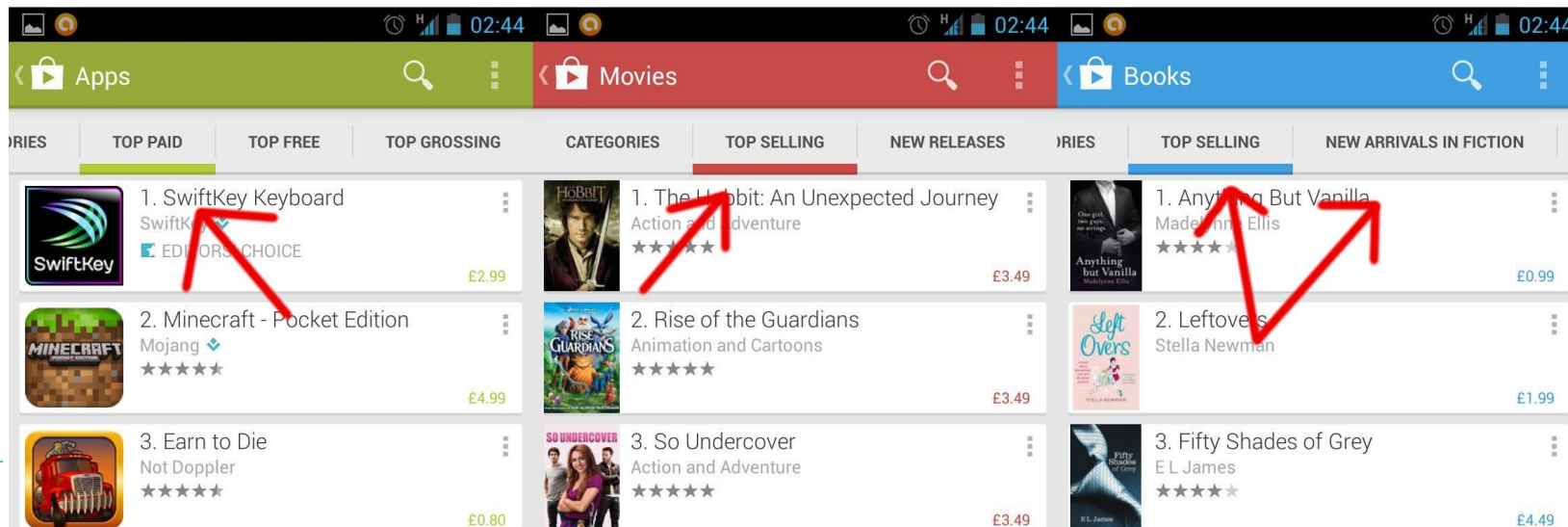
- ▶ Suited for showing data sets that represent themselves through **images or icons**.
- ▶ Unlike Lists Grid list may scroll either **vertically or horizontally**
- ▶ Rationale for using Grid-based Layout:
 - ▶ **Images/Icon can caught user attention more effectively**
 - ▶ In some situation, GridList can convey more meaning in images



Application Navigation: Tab



- ▶ Easy for users to explore and switch between views - browse **categorized data sets**
- ▶ Tab can either be **scrollable or fixed**
- ▶ Often used to enable user to **quickly navigate** between different part of your application





View Pager Navigation

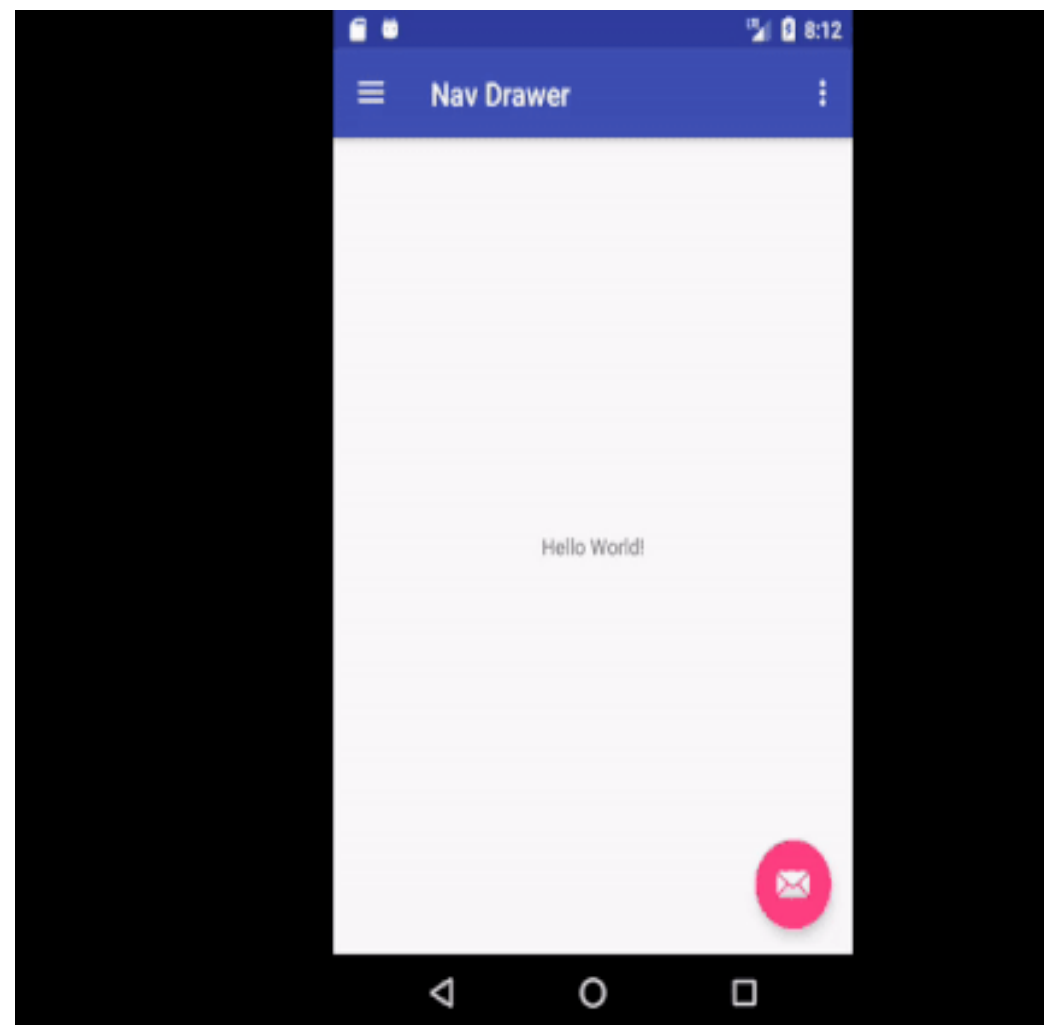
- ▶ Relies on **navigation indicator** (represented in circles or dots)
- ▶ Easy for users to explore and **switch between views** of application with a finger swipe
- ▶ Usually represents information in a **full screen view**





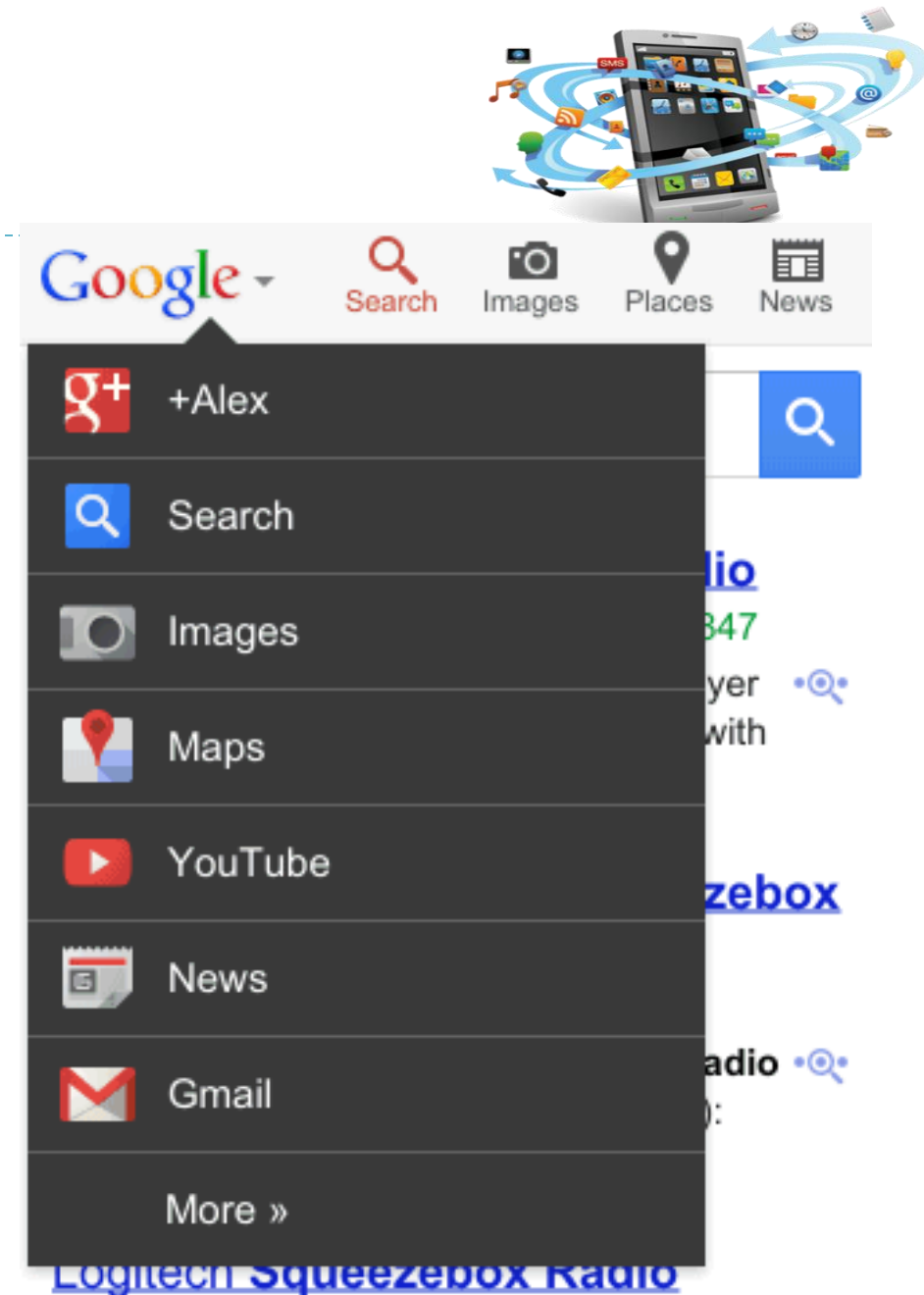
Application Navigation: Navigation Drawer

- ▶ A panel that **displays the app main navigation options on the left edge of the screen**
- ▶ It is **hidden** most of the time, but is revealed when the user swipes the finger from the left edge of the screen.
- ▶ **Does not consume spaces**
- ▶ Provides a **quick way for user to navigate from one view to another view** without having to press the “Back” button



Application Navigation: Drop Down

- ▶ Useful when changing the content is **important but not necessarily frequent** occurrence.
- ▶ More preferred where **switching the content is more frequent**
- ▶ **Do not change with the rest of screen**
- ▶ **Occupies small area** on the toolbar / action bar on an application.





End of Lecture

